

Name: _____

Date: _____

SURDS, (ADDING AND SUBTRACTING SURDS)

LO: I can add and subtract surds by collecting like surd terms.

Adding and subtracting surds

A **surd** is an irrational root such as $\sqrt{2}$ or $\sqrt{5}$. We can only **add** or **subtract surds** that have the same number under the root sign (like terms).

Simplify first: Sometimes surds need to be simplified before they can be combined. For example, $\sqrt{8} = 2\sqrt{2}$.

Quick examples

●	$3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$	like surds, add coefficients
●	$12\sqrt{3} = 2\sqrt{3} + 10\sqrt{3}$ $= 3\sqrt{3}$	simplify first
●	$7\sqrt{2} - 4\sqrt{2} = 3\sqrt{2}$	subtract coefficients
●	$3 + \sqrt{5} = \sqrt{5}$	cannot add unlike surds under one root
●	$2\sqrt{3} + 3\sqrt{2} = 5\sqrt{5}$	different surds cannot be combined

Key idea

Only like surds (same radicand) can be added or subtracted.
Simplify surds first to reveal like terms.

$$a\sqrt{n} + b\sqrt{n} = (a + b)\sqrt{n}$$

Collect like surd terms

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – like surds

Simplify $4\sqrt{3} + 7\sqrt{3}$.

Both terms have $\sqrt{3}$
Add coefficients: $4 + 7 = 11$
 $= 11\sqrt{3}$

11 $\sqrt{3}$

Treat surds like algebra: add the coefficients of like terms.

Example 2 – subtraction

Simplify $9\sqrt{2} - 5\sqrt{2}$.

Both terms have $\sqrt{2}$
Subtract coefficients: $9 - 5 = 4$
 $= 4\sqrt{2}$

4 $\sqrt{2}$

Same process for subtraction.

Example 3 – simplify first

Simplify $18 + \sqrt{2}$.

$18 = (9 \times 2) = 3\sqrt{2}$
 $3\sqrt{2} + \sqrt{2} = 4\sqrt{2}$
Coefficients: $3 + 1 = 4$

4 $\sqrt{2}$

Simplify surds before combining.

Example 4 – mixed expression

Simplify $50 - 8 + 32$.

$50 = 5\sqrt{2}$, $8 = 2\sqrt{2}$, $32 = 4\sqrt{2}$
 $5\sqrt{2} - 2\sqrt{2} + 4\sqrt{2}$
 $= 7\sqrt{2}$

7 $\sqrt{2}$

Simplify each surd, then combine like terms.

YOUR TURN – PRACTICE (deliberate practice)

1. Adding like surds

Simplify each expression.

- a) $2\sqrt{5} + 3\sqrt{5}$
- b) $6\sqrt{7} + \sqrt{7}$
- c) $4\sqrt{3} + 5\sqrt{3}$
- d) $11\sqrt{8} + 11\sqrt{8}$

2. Subtracting like surds

Simplify each expression.

- a) $8\sqrt{2} - 3\sqrt{2}$
- b) $10\sqrt{3} - 7\sqrt{3}$
- c) $5\sqrt{6} - 5\sqrt{6}$
- d) $12\sqrt{5} - 4\sqrt{5}$

3. Simplify first, then combine

Simplify each surd, then add or subtract.

- a) $12 + \sqrt{3}$
- b) $20 + \sqrt{5}$
- c) $27 - \sqrt{3}$
- d) $45 + \sqrt{20}$

4. Harder combinations

Simplify fully.

- a) $50 + \sqrt{18}$
- b) $48 - \sqrt{12}$
- c) $2\sqrt{8} + 3\sqrt{2}$
- d) $75 - \sqrt{27} + \sqrt{12}$

5. Mixed surd addition and subtraction

Simplify each expression fully.

- a) $3\sqrt{2} + \sqrt{8} - \sqrt{18}$
- b) $6\sqrt{3} + \sqrt{28} - \sqrt{7}$
- c) $2\sqrt{12} + \sqrt{48} - \sqrt{27}$
- d) $5\sqrt{3} - \sqrt{12} + \sqrt{75}$
- e) $98 - \sqrt{50} + \sqrt{32}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$ ☐

Correct answer: _____

Reason: _____

b) $3 + \sqrt{5} = 8$ ☐

Correct answer: _____

Reason: _____

c) $12 = 2\sqrt{3}$, so $12 + \sqrt{3} = 3\sqrt{3}$ ☐

Correct answer: _____

Reason: _____

d) $2\sqrt{3} + 3\sqrt{2} = 5\sqrt{6}$ ☐

Correct answer: _____

Reason: _____

e) $50 = 5\sqrt{2}$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Simplify $200 + \sqrt{50} - \sqrt{72} + \sqrt{18}$. Express your answer in the form $a\sqrt{2}$ where a is an integer.

Expression: _____

Simplified: _____

b) The sides of a triangle are 48 cm , 27 cm , and 75 cm . Find the perimeter in simplified surd form. If the perimeter is doubled, write the new perimeter.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

